MILITARY SPECIFICATION SHEET

PIN-RIVET, THREADED, TYPE III (100° FLUSH CROWN SHEAR HEAD), CLASS 6
(PH13-8 Mo CORROSION RESISTANT STEEL, 125 KSI Fsu AND 220 KSI Ftu)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the pin-rivet, threaded described herein shall consist of this document and the issue in effect of MIL-P-87988.

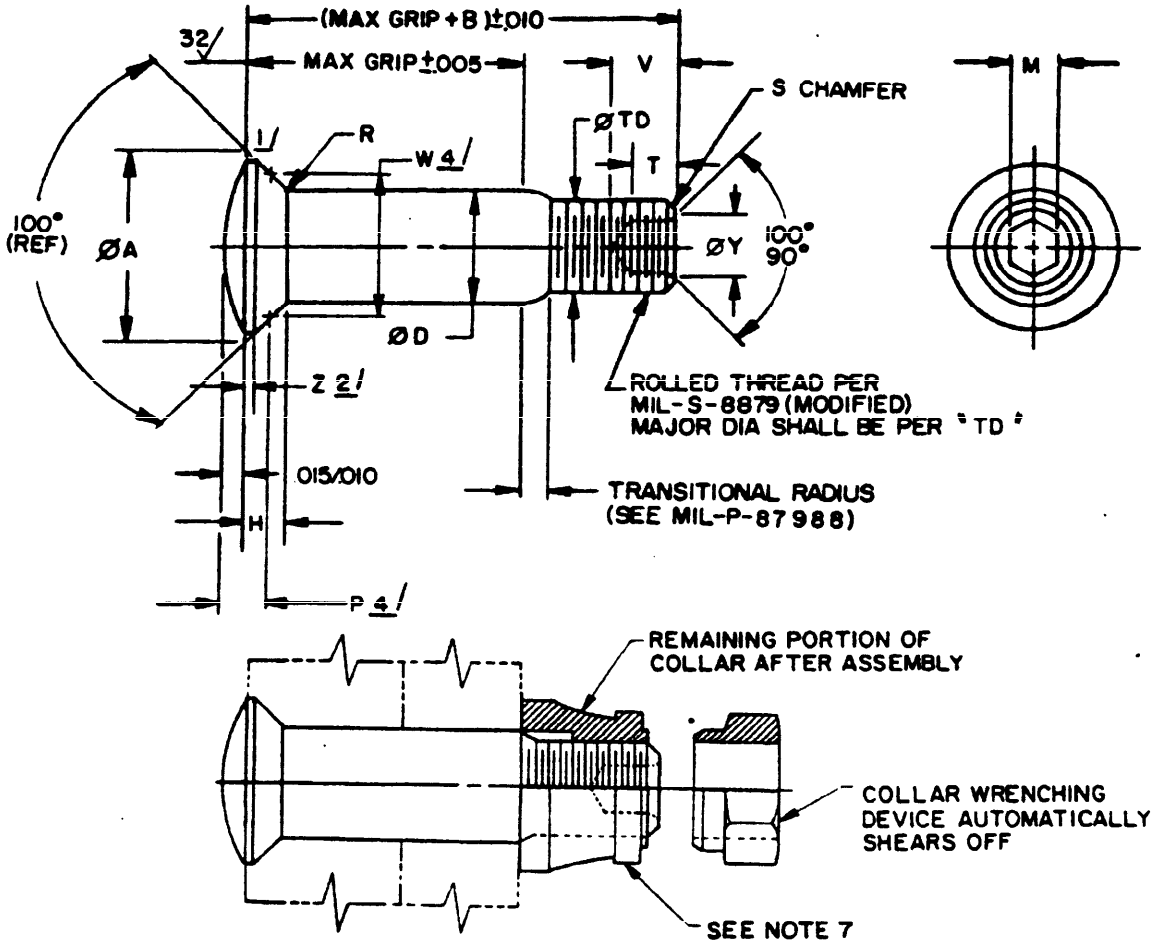


FIGURE 1. PIN-RIVET, THREADED, TYPE III (100° FLUSH CROWN SHEAR HEAD), CLASS 6 (PH13-8 Mo CORROSION RESISTANT STEEL, 125 KSI Fsu AND 220 KSI Ftu)

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NOTES:

1. Dimensions in inches to be met after application of final coating or surface finish, unless otherwise specified.
1/ 2. Head edge out of roundness shall not exceed "F".
3. Concentricity: conical surface of head to "D" diameter within .005 FIM.
4. "H" dimensioned from maximum "D" diameter.
3/ 5. Evidence of broken edge across points.
2/ 6. Curved or flat edge manufacturer's option.
7. While this part is designed to be used in interference fit applications (because it has a transitional radius) it can also be used in noninterference or clearance applications. It is similar to HLT51. "HL" is a registered trademark of Hi-Shear Corporation.
8. Hole preparation in accordance with NAS618. (Information only) .
4/ 9. All head dimensions shall comply with requirements of TABLE I of this slash sheet and TABLE X of procurement specification.

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TABLE I. DASH NUMBERS AND DIMENSIONS

DIA NO.	NOM SIZE	THREAD (MODIFIED)	ØA	B REF	ØD WITHOUT SOLID FILM LUBE	ØD WITH SOLID FILM LUBE	ØTD	1/ F	H	R RAD	2/ Z MAX	S CHAMFER REF	SOCKET M HEX	SOCKET T DEPTH	SOCKET V MAX DRILL DEPTH	SOCKET ØY	DOUBLE SHEAR POUNDS MIN	TENSION POUNDS MIN
5	5/32	.1640-32 UNJC-3A	.2612 .2564	.312	.1635 .1630	.1635 .1625	.1595 .1570	.004	.0408 .0388	.025 .015	.010	.031 x 45°	.0801 .0791	.100 .080	.140	3/	5,280	1,700
6	3/16	.1900-32 UNJF-3A	.3016 .2966	.325	.1895 .1890	.1895 .1885	.1840 .1810	.005	.0470 .0449	.030 .020	.015	.031 x 45°	.0806 .0791	.100 .080	.145	.119 .104	7,060	2,600
8	1/4	.2500-28 UNJF-3A	.3948 .3898	.395	.2495 .2490	.2495 .2485	.2440 .2410	.006	.0610 .0589	.030 .020	.015	.031 x 45°	.0967 .0947	.110 .090	.160	.142 .122	12,260	4,400
10	5/16	.3125-24 UNJF-3A	.4739 .4689	.500	.3120 .3115	.3120 .3110	.3060 .3020	.007	.0679 .0658	.040 .030	.015	.047 x 45°	.1295 .1270	.130 .110	.200	.180 .160	19,160	7,000
12	3/8	.3750-24 UNJF-3A	.5604 .5554	.545	.3745 .3740	.3745 .3735	.3680 .3640	.008	.0780 .0759	.040 .030	.015	.047 x 45°	.1617 .1582	.160 .140	.239	.217 .197	27,600	10,000
14	7/16	.4375-20 UNJF-3A	.6680 .6620	.635	.4370 .4365	.4370 .4360	.4310 .4260	.009	.0969 .0944	.050 .040	.022	.047 x 45°	.1930 .1895	.190 .170	.278	.253 .233	37,500	12,500
16	1/2	.5000-20 UNJF-3A	.7540 .7480	.685	.4995 .4990	.4995 .4985	.4930 .4880	.010	.1068 .1043	.050 .040	.022	.047 x 45°	.2242 .2207	.220 .200	.323	.289 .269	49,100	18,000

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REQUIREMENTS:

1. Material. PH13-8Mo stainless steel (UNS S13800) in accordance with AMS 5629.

2. Protective coating and surface treatment

All pins shall be passivated in accordance with QQ-P-35.

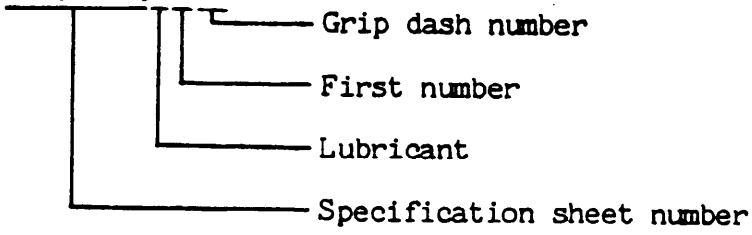
Lubricant:
B - Solid film lubricant in accordance with MIL-L-46010 Type I.
C - Cetyl alcohol lubricant in accordance with MIL-L-87132 Type I, Grade B.

3. Heat treatment. 125,000 psi shear minimum in accordance with MIL-H-6875.

4. Surface texture. Unless otherwise specified, all surfaces shall have a 125 microinch finish.

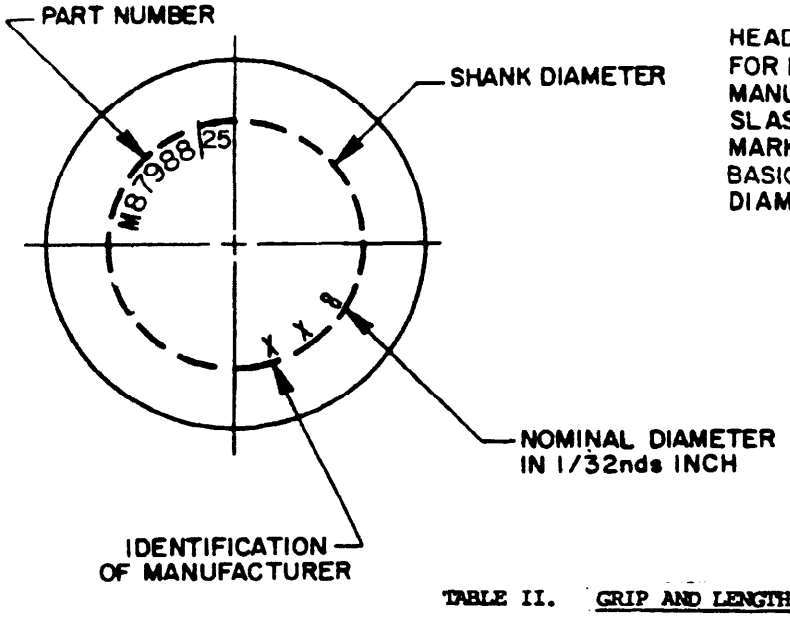
5. Part number. The part number consists of the letter "M", the basic number of this specification sheet, followed by a lubricant code letter, the first number indicates nominal diameters in thirty-seconds of an inch taken from TABLE I, the grip dash number indicates maximum grip length in sixteenths of an inch taken from TABLE II.

Example: M87988/25B8-8



M87988/25B8-8 indicates: Pin-Rivet, Threaded, Type III (100° Flush Crown Shear Head), Class 6 (PH13-8 Mo Corrosion Resistant Steel, 125 KSI Fsu and 220 KSI Ftu); passivated, solid film lubricant; 8/32 nominal diameter; 8/16 maximum grip length.

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HEAD MARKING SHALL BE .010 MAXIMUM DEEP. FOR DASH 5,6 AND 8 SIZES ONLY, MARK MANUFACTURER IDENTIFICATION AND M PLUS SLASH SHEET (M25). FOR DASH 10 AND ABOVE, MARK MANUFACTURER IDENTIFICATION PLUS BASIC PART NUMBER (M87988/25-)AND DIAMETER.

TABLE II. GRIP AND LENGTH DIMENSIONS

GRIP DASH NO.	MAX GRIP ±.005	LENGTH ±.010[a/]						
		DIAMETER DASH NUMBER AND THREAD SIZE						
		5 .1640-32	6 .1900-32	8 .2500-28	10 .3125-24	12 .3750-24	14 .4375-20	16 .5000-20
1	.062	.375						
2	.125	.437	.450	.520	.625	.670	.760	.810
3	.188	.500	.513	.583	.688	.733	.823	.873
4	.250	.562	.575	.645	.750	.795	.885	.935
5	.312	.625	.638	.708	.813	.858	.948	.998
6	.375	.687	.700	.770	.875	.920	1.010	1.060
7	.438	.750	.763	.883	.938	.984	1.073	1.123
8	.500	.812	.825	.895	1.000	1.045	1.135	1.185
9	.562	.875	.888	.958	1.063	1.108	1.198	1.248
10	.625	.937	.950	1.020	1.125	1.170	1.260	1.310
11	.688	1.000	1.013	1.083	1.188	1.233	1.323	1.373
12	.750	1.062	1.075	1.145	1.250	1.295	1.385	1.435
13	.812	1.125	1.138	1.208	1.313	1.358	1.448	1.498
14	.875	1.187	1.200	1.270	1.375	1.420	1.510	1.560
15	.938	1.250	1.263	1.333	1.438	1.483	1.573	1.623
16	1.000	1.312	1.325	1.395	1.500	1.545	1.635	1.685
17	1.062	1.375	1.388	1.458	1.563	1.608	1.698	1.748
18	1.125	1.437	1.450	1.520	1.625	1.670	1.760	1.810
19	1.188	1.500	1.513	1.583	1.688	1.733	1.823	1.873
20	1.250	1.562	1.575	1.645	1.750	1.795	1.885	1.935
21	1.312	1.625	1.638	1.708	1.813	1.858	1.948	1.998
22	1.375	1.687	1.700	1.770	1.875	1.920	2.010	2.060
23	1.438	1.750	1.763	1.833	1.938	1.983	2.073	2.123
24	1.500	1.812	1.825	1.895	2.000	2.045	2.135	2.185
25	1.562	1.875	1.888	1.958	2.063	2.108	2.198	2.248
26	1.625	1.937	1.950	2.020	2.125	2.170	2.260	2.310
27	1.688	2.000	2.013	2.083	2.188	2.233	2.323	2.373
28	1.750	2.062	2.075	2.145	2.250	2.295	2.385	2.435
29	1.812	2.125	2.138	2.208	2.313	2.358	2.448	2.498
30	1.875	2.187	2.200	2.270	2.375	2.420	2.510	2.560
31	1.938	2.250	2.263	2.333	2.438	2.483	2.573	2.623
32	2.000	2.312	2.325	2.395	2.500	2.545	2.635	2.685

a/ Nominal length equals maximum grip plus "B". See Figure 1.

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Army - AV
Navy - AS
Air Force - 99

Review activities:
Army - GL
DLA - IS
NSA - NS

User activities:

Preparing activity:
Air Force - 82

Agent:
DLA - IS

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